ADSP 31014 Statistical Models for Data Science
Homework 3

(25 points) Question 1a
Two Categorical Inputs: Let's continue with our `chicago_taxi_trip_sample.csv` data set. In our lecture, we predict Chicago taxi trip trip_seconds based on the two explanatory variables fare and hour_type along with their interaction. Now let's consider pickup_area as another explanatory variable.

- Fit a linear regression model with trip_seconds as the response variable and both hour_type and pickup_area as explanatory variables (without any interaction) i.e.

```
lm(trip_seconds ~ hour_type + pickup_area, ...)
```

- What is the R^2 value of the fitted model?
- How many β's parameters does this model have?
- Write out the fitted model equation in the mathematical form

$$y = \beta_0^* + \beta_1^* \mathbf{1}_{\{\text{hour_type}=\cdots\}} + \cdots + \beta_?^* \mathbf{1}_{\{\text{pickup_area}=\cdots\}} + \cdots$$

- What is the predicted value $\hat{y}$ when `hour_type = Rush Hour` and `pickup_area = Downtown`?

(40 points) Question 1b
Categorical Interaction: Continuing from Question 1a, let's also consider the interaction between hour_type and pickup_area.

- Fit a linear regression model with trip_seconds as the response variable and both hour_type and pickup_area as explanatory variables along with their interaction i.e.

```
lm(trip_seconds ~ hour_type * pickup_area, ...)
```

- What is the R^2 value of the fitted model?
- How many β's parameters does this model have?
- What is the predicted value $\hat{y}$ when `hour_type = Rush Hour` and `pickup_area = O'Hare`?
- For each combination of the hour_type and pickup_area values, compute the sample mean of trip_seconds from those subset of rows in the data set. (Hint: In R, use `tapply()` or `dplyr` package's `group_by()` and `summarize()`. In Python, use Pandas's `groupby()`.)
- Which combination of hour_type and pickup_area values has the smallest sample mean of trip_seconds? What is the value of this sample mean?
- What is the relationship between these sample means vs the fitted β's coefficients of the model?
- Based on the numbers of levels of the two categorical explanatory variables, explain why the number of β's parameters of this model make sense?

(30 points) Question 1c
Include Fare: So far we omitted the useful fare explanatory variable. Let's include it now.

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- Fit a linear regression model with `trip_seconds` as the response variable and both `hour_type` and `pickup_area` as explanatory variables along with their interaction, and also the `fare` explanatory variable as follows:

  ```
  lm(trip_seconds ~ hour_type * pickup_area + fare:(hour_type * pickup_area), ...)
  ```

- What is the R^2 value of the fitted model?
- How many β's parameters does this model have?
- This model essentially fits a separate linear regression (i.e. straight line) to `trip_seconds` vs `fare` for each subset of rows of data per `hour_type` and `pickup_area` combination. Based on this fact, explain why the number of β's parameters of this model make sense?
- For the combination of `hour_type = Rush Hour` and `pickup_area = O'Hare`, the overall model equation implies a specific linear equation (i.e. straight line) for this case in the form

  $$y = a_0 + a_1(\text{fare})$$

  Based on the model summary i.e. `summary(lm(...))`, what are the value of a_0 and a_1?
- Verify your answer of a_0 and a_1 by fitting a separate linear regression model to the subset of rows in which `hour_type = Rush Hour` and `pickup_area = O'Hare`.

(10 points) Question 1d
Three-Way Interaction: The R's model formula in Question 1c looks a bit weird and long. Let's try another way.

- Fit a linear regression model with `trip_seconds` as the response variable as follows:

  ```
  lm(trip_seconds ~ hour_type * pickup_area * fare, ...)
  ```

- Based on the model summaries, how is this model related to the model in Question 1c?

(5 points) Question 2
Ask some GenAI tool (e.g. ChatGPT, Gemini, Claude) about how to handle a categorical variable with many levels when fitting a linear regression model. What does GenAI say?

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